

Summary of Backpropagation-Free Training

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1 Model Accuracy: Add Noise to the Activation Value

Table 1: Model accuracy with different noise addition method.

Loss Computation Strategy	Model	Accuracy (%)
Published results	ResNet-5	64.8
	VGG-8	77.3
Experimental results	ResNet-5	63.53
	VGG-8	71.63

In the open-source implementation of [1], both the VGG-8 and ResNet-5 models add noise to the model parameters in the first layer. However, this approach is incompatible with our proposed model splitting architecture. To address this issue, I modified the first layer by adding noise to the activation values instead of the parameters. Table 1 compares the accuracy of our modified models with the results reported in [1].

2 Model Splitting

I have successfully implemented a BP-free model splitting framework. In this newly proposed framework, the client is only required to perform a single forward pass during each training iteration, effectively reducing computational and memory overhead by approximately 50%. Furthermore, the communication cost between the client and the server is reduced by about 40% compared to traditional BP-based methods.

However, this framework currently exhibits mixed impacts on model accuracy depending on the architecture. One observed issue stems from the optimizer configuration in PyTorch. Specifically, since the client and server models are updated using separate optimizers, their update strategies deviate from those used in centralized training. This discrepancy can lead to instability in model convergence and ultimately affects the overall performance. For instance, under this distributed setup, VGG-8 achieves an improved test accuracy of 73.37%, up from 71.63% in centralized training, whereas ResNet-5 experiences a drop in accuracy, decreasing from 63.53% to 57.05%.

3 Larger Model

To achieve higher accuracy, I attempted to apply larger models within the BP-free framework. Previously, ResNet18 and VGG16 were explored; however, these models proved too large to be trained on our GPU using the BP-free approach. In this attempt, I adopted a different strategy: building upon the open-source VGG8 and ResNet5 models introduced in [1], I increased their depth by adding additional layers, aiming to improve model capacity and accuracy.

Despite this architectural enhancement, the resulting accuracy was lower than expected. The main challenge stemmed from the difficulty in determining appropriate noise iteration settings for each layer

in the BP-free framework.

Initially, I assumed that increasing the number of iterations per layer would yield better results. However, experimental results disproved this assumption, showing that an excessively large number of iterations can actually degrade performance. The open-source implementation accompanying [1] appears to provide empirically optimized iteration settings for its original models. However, when scaling up the model depth, it becomes unclear how to adjust these settings accordingly.

To address this issue, I plan to communicate with the authors of [1] to gain further insights into the principles behind their parameter choices and explore potential strategies for extending them to deeper architectures.

References

- [1] J. Jiang, Z. Zhang, C. Xu, Z. Yu, and Y. Peng, “One forward is enough for neural network training via likelihood ratio method,” *arXiv preprint arXiv:2305.08960*, 2023.